Motion in a Central Potential; the Kepler Problem — approved segments

10 approved segments, in reading order

CHAPTER 3

MOTION IN CENTRAL POTENTIAL $V(r)$

$$\vec{L} = \mu r^2 \dot{\phi}\, \hat{z} \qquad \vec{r} = r\,\hat{r} \qquad \vec{p} = \mu \dot{\vec{r}}$$

CONST OF MOTION

$$L = T - V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - V(r)$$

$$\phi \text{ CYCLIC} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Rightarrow p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \mu r^2 \dot{\phi} \equiv \ell$$

CONSERVED

$$V_{eff}(r) = V(r) + \frac{\ell^2}{2\mu r^2}$$

$$E = T + V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = \frac{\mu}{2}\dot{r}^2 + V_{eff}(r)$$

UNIVERSAL METHOD $\Rightarrow$ COMPLETE SOLUTION

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$$V_{eff}(r) = V(r) + \frac{\quad}{2\mu r^2}$$

$$E = T + V = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = \frac{\mu}{2}\dot{r}^2 + V_{eff}(r)$$

UNIVERSAL METHOD $\Rightarrow$ COMPLETE SOLUTION

$$\boxed{\begin{aligned} t &= \int_{r_0}^{r} \frac{dr'}{\sqrt{\frac{2}{\mu}\left(E - V(r') - \frac{\ell^2}{2\mu r'^2}\right)}} \\ \phi &= \phi_0 + \int_{t}^{t} \frac{\ell}{\mu r^2(t')}\, dt' \end{aligned}}$$

$$t = t(r)$$

$$\phi = \phi(t)$$

$$\mu\ddot{r} =$$

$$\frac{\partial}{\partial r}($$

LHS:

LHS:

$$\mu\ddot{r} = -\frac{\ell^2}{\mu}\frac{1}{r^2}\frac{d}{d\phi}\frac{du}{d\phi} = -\frac{\ell^2}{\mu}u^2\frac{d^2u}{d\phi^2}$$

LHS=RHS

$$-\frac{\ell^2}{\mu}u^2$$

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TENTIAL $V(r)$

$-\hat{r}$ $\quad \vec{p} = \mu \dot{\vec{r}}$

$-V(r)$

$= \frac{\partial L}{\partial \dot\phi} = \mu r^2 \dot\phi \equiv l$

ONSERVED

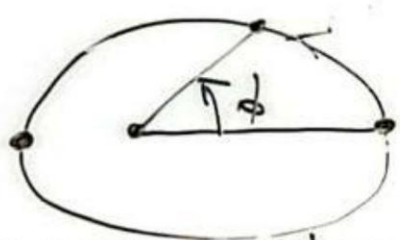
$(r) = \frac{\mu}{2}\dot{r}^2 + V_{eff}(r)$

PLETE SOLUTION

DIFF EQN FOR $r = r(\phi)$

$$l = \mu r^2 \dot\phi \Rightarrow \dot\phi = \frac{l}{\mu r^2} = \frac{d\phi}{dt}$$

$$\frac{df}{dt} = \frac{df}{d\phi}\frac{d\phi}{dt} = \underbrace{\frac{l}{\mu r^2}\frac{df}{d\phi}}$$



$$-\frac{\partial V}{\partial r} = F(r)$$

Radial Eq of Motion

$$\mu \ddot{r} = -\frac{\partial V_{eff}}{\partial r} = -\frac{\partial}{\partial r}\left(V(r) + \frac{l^2}{2\mu r^2}\right)$$

$$F(r) = -\frac{\partial V}{\partial r}$$

$$\frac{dV}{dr} = \frac{dV}{du}\frac{du}{dr}$$

$$\frac{du}{dr} = -\frac{1}{r^2} = -u^2$$

$$F(1/u) = F(r) = -\frac{dV}{dr} = u^2$$

$$-F(1/u) = -u^2\, dV(1/u)/du$$

$$-F\left(\frac{1}{u}\right) = \frac{l^2}{\mu} u^2 \Big[$$

$$-\cancel{u^2}\frac{dV}{du} = \frac{l^2}{\mu}\cancel{u^2}\Big[$$

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$V(r) = \frac{\mu}{2}\dot{r}^2 + V_{eff}(r)$

PLETE SOLUTION

$t = t(r)$

$\phi = \phi(t)$

$\frac{d^2u}{d\phi^2}$

Radial Eq of Motion

$$\mu\ddot{r} = -\frac{\partial V_{eff}}{\partial r} = -\frac{\partial}{\partial r}\left(V(r) + \frac{l^2}{2\mu r^2}\right)$$

$$-\frac{\partial V}{\partial r} = F(r)$$

$$-F\left(\frac{1}{u}\right) = \frac{l}{\mu}u^2\Big[$$

$$-\cancel{u^2}\frac{dV}{du} = \frac{l^2}{\cancel{\mu}}u^2\Big[$$

$$\mu\ddot{r} = F(r) + \frac{l^2}{\mu r^3} \qquad u = \frac{1}{r} \qquad \frac{du}{d\phi} = \frac{du}{dr}\frac{dr}{d\phi} = -\frac{1}{r^2}\frac{dr}{d\phi}$$

$$\frac{\partial}{\partial r}\left(r^{-2}\right) = (-2)r^{-3} \qquad \frac{l^2}{\mu r^3} = \frac{l^2}{\mu}u^3 \Rightarrow \text{RHS}: F(1/u) + \frac{l^2}{\mu}u^3$$

$$\text{LHS}: \mu\frac{d}{dt}\frac{dr}{dt} = \mu\frac{d}{dt}\frac{l}{\mu r^2}\frac{dr}{d\phi} = \cancel{\mu}\frac{l}{\cancel{\mu} r^2}\frac{d}{d\phi}\left(\frac{l}{\mu r^2}\frac{dr}{d\phi}\right)$$

LHS=RHS

$$\boxed{-\frac{l^2}{\mu}u^2\frac{d^2u}{d\phi^2} = F(1/u) + \frac{l^2}{\mu}u^3}$$

GENERIC EQ. OF TRAJECTORY FOR ARBITRARY $V(r)$

EQ

SOL

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$$F(r) = -\frac{\partial V}{\partial r}$$

$$\frac{dV}{dr} = \frac{dV}{du}\,\frac{du}{dr} = -u^2\,\frac{dV}{du}$$

$$\frac{du}{dr} = -\frac{1}{r^2} = -u^2$$

$$F(1/u) = F(r) = -\frac{dV}{dr} = u^2\,\frac{dV(1/u)}{du}$$

$$-F(1/u) = -u^2\, dV(1/u)/du$$

$$F(r) \qquad -F\left(\tfrac{1}{u}\right) = \frac{\ell^2}{\mu}u^2\left[u + \frac{d^2u}{d\phi^2}\right]$$

$$-\cancel{u^2}\,\frac{dV}{du} = \frac{\ell^2}{\mu}\cancel{u^2}\left[u + \frac{d^2u}{d\phi^2}\right]$$

$$\frac{\ell^2}{2\mu r^2}\Big)$$

GRAVITATI

KEPL

$$V = -\frac{k}{r} = -k$$

$$\boxed{\frac{d^2u}{d\phi^2} + u =}$$

ASIDE

$$\tilde{u} = u - \frac{k\mu}{\ell^2}$$

$$\boxed{\frac{d^2\tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

$$u(\phi) - k\mu$$

$$F(r) = \ldots \left(\frac{1}{u}\right) - \frac{\ell^2}{\mu} u^2 \left[u + \frac{d^2u}{d\phi^2} \right]$$

$$\ldots \frac{\ell^2}{2\mu r^2} \Big) \quad - \cancel{u^2} \frac{dV}{du} = \frac{\ell^2}{\mu} \cancel{u^2} \left[u + \frac{d^2u}{d\phi^2} \right]$$

$$\frac{du}{d\phi} = \frac{du}{dr}\frac{dr}{d\phi} = \frac{-1}{r^2}\frac{dr}{d\phi}$$

RHS: $F(1/u) + \frac{\ell^2}{\mu} u^3$

$$\frac{\ell}{\mu r^2}\frac{d}{d\phi}\left(\frac{\ell}{\mu r^2}\frac{dr}{d\phi}\right)$$

GENERIC EQ. OF TRAJECTORY FOR ARBITRARY $V(r)$

$$\boxed{\frac{d^2u}{d\phi^2} + u = -\frac{\mu}{\ell^2}\frac{d}{du}V(1/u)}$$

EQ FOR TRAJECTORY $r(\phi)$

SOLVE FOR GIVEN $u = u_0$ FOR $\phi = \phi_0$ $\quad \frac{du}{d\phi} = 0$

$\frac{du}{d\phi} = 0 \Leftrightarrow$ MAX OR MIN r POINT

$$\boxed{\frac{d^2\tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

$$u(\phi) = \frac{k\mu}{\ell^2} \ldots$$

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$-u^2 \frac{dV}{du}$

$\frac{dV(1/u)}{du}$

$+ \frac{d^2u}{d\phi^2}$

$\frac{d^2u}{d\phi^2}$

GRAVITATION CASE

KEPLER PROBLEM

$$V = -\frac{k}{r} = -ku \qquad \frac{dV}{du} = -k$$

$$\boxed{\frac{d^2u}{d\phi^2} + u = \frac{k\mu}{\ell^2}}$$

HARMONIC OSCILLATOR WITH SHIFTED ORIGIN

ASIDE

$$\tilde{u} = u - \frac{k\mu}{\ell^2}$$

$$\boxed{\frac{d^2\tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

HOMOGENEOUS SOLUTION

$$\tilde{u}(\phi) = A\cos(\phi - \phi_0)$$

$$\tilde{u}'(\phi) = -A\sin(\phi - \phi_0)$$

$$\tilde{u}''(\phi) = -A\cos(\phi - \phi_0) = -\tilde{u}(\phi)$$

$$u(\phi) = \frac{k\mu}{\ell^2} + A\cos(\phi - \phi_0)$$

$u(\phi) =$

e co

USING

$u'' =$

$\frac{du}{d\phi} = (u')^2$

$d\phi$

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$d\phi^2 \big]$

$\frac{d^2 u}{d\phi^2} \big]$

$$\boxed{\frac{d^2\tilde{u}}{d\phi^2} + \tilde{u} = 0}$$

$$\tilde{u}'(\phi) = -A\sin(\phi-\phi_0)$$

$$\tilde{u}''(\phi) = -A\cos(\phi-\phi_0) = -\tilde{u}(\phi)$$

$$\boxed{u(\phi) = \frac{k\mu}{l^2} + A\cos(\phi-\phi_0)}$$

$$\boxed{= -\frac{\mu}{l^2}\frac{d}{du}V(1/u)}$$

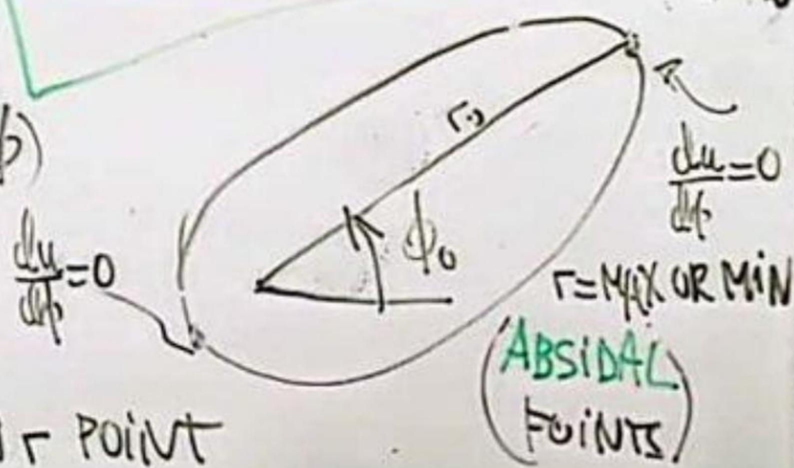
TRAJECTORY $r(\phi)$

FOR GIVEN

u_0 FOR $\phi=\phi_0$

$=0 \Leftrightarrow$ MAX OR MIN r POINT

$u_0 = \frac{1}{r_0}$



$\frac{du}{d\phi} = (u')^2$

$d\phi$

$u = \frac{\mu}{l}$

$e =$

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SE
OBLEM

$= -k$

HARMONIC
OSCILLATOR
ITH SHIFTED
ORIGIN

OUS SOLUTION

$\cos(\phi - \phi_0)$

$A\sin(\phi - \phi_0)$

$\cos(\phi - \phi_0)$

$$\boxed{u(\phi) = \frac{k\mu}{\ell^2}\left[e\cos(\phi - \phi_0) + 1\right] = \frac{1}{r}}$$

e CONSTANT $\quad e \equiv \frac{A}{k\mu/\ell^2}$

USING THE UNIVERSAL METHOD

$$u'' = -u + \frac{k\mu}{\ell^2}$$

$$\Downarrow \times u'(\phi)$$

$$\frac{du}{d\phi} = (u')^2 = \sqrt{\cdot \quad -}$$

$$d\phi = \frac{du}{}$$

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$\cos(\phi-\phi_0)$

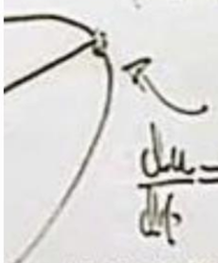
$A\sin(\phi-\phi_0)$

$\cos(\phi-\phi_0)$

$\phi)$

$\phi_0)$

$u_0 = \frac{1}{r_0}$



$\frac{du}{d\phi}=0$

$r = $ MAX OR MIN

(APSIDAL POINTS)

$\Downarrow \times u'(\phi)$

$\frac{du}{d\phi} = (u')^2 = \sqrt{\;\cdot\;\;-\;}$

$d\phi = \frac{du}{u'}$

$\vdots$

$$\boxed{u = \frac{\mu k}{\ell^2}\left[1 + \sqrt{1+\frac{2\ell^2 E}{\mu k^2}}\cos(\phi-\phi_0)\right] = \frac{1}{r}}$$

$$\boxed{e \equiv \sqrt{1+\frac{2\ell^2 E}{\mu k^2}}}$$

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